

MongoDB Compass and Atlas Assignment

Local database: **spotifyDB** | Atlas database: **instaClone** | Collection: **zomatoRestaurants**

Name: _____ Roll No.: _____ Date: _____

Note: These are the steps and the expected results. Menu names in Compass and Atlas can look slightly different between versions. If your teacher wants proof, take a screenshot after each question and paste it under the answer. The commands in Question 4 are run in a normal terminal (Command Prompt, PowerShell or Terminal), not inside mongosh.

Question 1: Install Compass, connect, and create spotifyDB with playlists

Install MongoDB Compass, connect to your local MongoDB server, and create a database named spotifyDB with a collection called playlists.

Steps

1. Download MongoDB Compass from the official MongoDB website (Products > Tools > Compass) and install it (.exe or .msi on Windows, .dmg on macOS, .deb or .rpm on Linux).
2. Make sure the local MongoDB server is running (the mongod service on the default port 27017).
3. Open Compass, click **+ Add new connection**, enter the connection string below and click **Connect**.
4. In the left sidebar, click **Create database** (the + icon). Enter Database Name **spotifyDB** and Collection Name **playlists**, then click **Create Database**.

```
mongodb://localhost:27017
```

Result: spotifyDB now appears in the sidebar with an empty playlists collection inside it. MongoDB only keeps a database once it has at least one collection, which is why both names are entered together.

Question 2: Import restaurant data into zomatoRestaurants

Using Compass, import a JSON file with at least 10 restaurant documents (name, cuisine, rating) into a new collection named zomatoRestaurants in spotifyDB.

Step 1: Create a file named zomatoRestaurants.json with this data (12 documents)

```
[
  { "name": "Spice Route",      "cuisine": "Indian",      "rating": 4.5 },
  { "name": "Dragon Wok",      "cuisine": "Chinese",    "rating": 4.3 },
  { "name": "Bella Napoli",    "cuisine": "Italian",    "rating": 4.6 },
  { "name": "Thali House",     "cuisine": "Gujarati",   "rating": 4.8 },
  { "name": "Taco Fiesta",     "cuisine": "Mexican",    "rating": 3.9 },
  { "name": "Pizza Palace",    "cuisine": "Pizza",      "rating": 4.4 },
  { "name": "Biryani Junction", "cuisine": "Hyderabadi",  "rating": 4.2 },
  { "name": "Sushi Zen",       "cuisine": "Japanese",   "rating": 4.1 },
  { "name": "Dosa Corner",     "cuisine": "South Indian", "rating": 4.0 },
  { "name": "Kebab Mahal",     "cuisine": "Mughlai",    "rating": 4.3 },
  { "name": "Cafe Mocha",      "cuisine": "Cafe",       "rating": 3.8 },
  { "name": "Burger Barn",     "cuisine": "Fast Food",  "rating": 3.7 }
]
```

Step 2: Import it in Compass

1. Open spotifyDB in the sidebar and click **Create collection**. Name it **zomatoRestaurants**.
2. Open the zomatoRestaurants collection, click **ADD DATA** and choose **Import JSON or CSV file**.

3. Select `zomatoRestaurants.json`, keep the file type as **JSON**, and click **Import**.

Result: Compass shows a success message and the Documents tab shows **1 - 12 of 12**. Each document gets an automatic `_id` field.

Question 3: Create an Atlas account, a cluster, and the `instaClone` database

Sign up for a free MongoDB Atlas account, create a new cluster, and deploy a database named `instaClone`.

Steps

1. Go to the MongoDB Atlas website and sign up with an email address or a Google account. Verify your email.
2. Click **Create** (or **Build a Database**) and choose the free **M0** tier. Pick a cloud provider and a nearby region (for example Mumbai, if offered), give the cluster a name such as **Cluster0**, and create it. It takes a few minutes to deploy.
3. Under **Database Access**, add a database user (for example **atlasUser**) with a password and the 'Read and write to any database' privilege.
4. Under **Network Access**, click **Add IP Address** and add your current IP address. (For short practice work only, `0.0.0.0/0` allows all IPs, but it is not safe for real projects.)
5. Open your cluster, go to **Browse Collections** (Data Explorer) and click **Add My Own Data / Create Database**. Enter Database Name **instaClone** and a first Collection Name (for example **zomatoRestaurants**), then create it.

Result: The Atlas cluster is running and `instaClone` is listed under it. Atlas needs a collection name at the time of creating a database, which is why one is entered in the last step.

Question 4: Export with `mongoexport` and import into Atlas with `mongoimport`

Export the `zomatoRestaurants` collection from the local MongoDB using `mongoexport`, then import it into the `instaClone` database on Atlas using `mongoimport`.

Before you start

`mongoexport` and `mongoimport` come with the **MongoDB Database Tools**, which are a separate download from the MongoDB website. After installing, check with `mongoexport --version`.

Step 1: Export from the local server

```
mongoexport --db=spotifyDB --collection=zomatoRestaurants --out=zomatoRestaurants.json
```

Expected output:

```
connected to: mongodb://localhost/
exported 12 records
```

Step 2: Get the Atlas connection string

In Atlas, click **Connect** on your cluster, choose **Drivers** (or **Shell**) and copy the connection string. It looks like the example below. Replace `<user>` and `<password>` with your database user's details (if the password has special characters such as `@` or `#`, URL-encode them).

```
mongodb+srv://<user>:<password>@cluster0.abcde.mongodb.net/
```

Step 3: Import into instaClone on Atlas (type as one line)

```
mongoimport --uri "mongodb+srv://<user>:<password>@cluster0.abcde.mongodb.net/instaClone"
--collection zomatoRestaurants --file zomatoRestaurants.json
```

Expected output:

```
connected to: mongodb+srv://[**REDACTED**]@cluster0.abcde.mongodb.net/instaClone
12 document(s) imported successfully. 0 document(s) failed to import.
```

Explanation: mongoexport writes each document as one line of JSON, keeping the original `_id` values. mongoimport reads that file and inserts the documents into the collection named in the command, in the database given at the end of the Atlas connection string.

Question 5: Verify the import in Atlas using Compass

Connect to the Atlas instaClone database using Compass and verify that all restaurant documents were imported by viewing the zomatoRestaurants collection and counting the documents.

Steps

1. Open Compass and click **+ Add new connection**. Paste the Atlas connection string (with your user and password filled in) and click **Connect**.
2. In the sidebar, open **instaClone** and click **zomatoRestaurants**.
3. On the Documents tab, read the count shown at the top right, which reads **1 - 12 of 12**.
4. For a second check, open the **MONGOSH** tab at the bottom of Compass and run the command below.

```
db.zomatoRestaurants.countDocuments()
```

Expected output:

```
12
```

Conclusion: The count on Atlas is 12, the same as the local spotifyDB.zomatoRestaurants collection and the number reported by mongoimport, so all restaurant documents were imported successfully.